

PA14 Prevalence, impairment patterns, and response to cholinergic modulation of visual attention deficits in post-stroke aphasia

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INGRODUCTION

Visual attention deficits (VADs) have long been overlooked in people with aphasia (PWA) until recently, especially due to difficulties of assessment^{1,2}. However, recent evidence highlights their influence on language outcomes^{3,4,5,6}. Yet, despite improvements in attention rehabilitation for this population⁷, no pharmacological clinical trials have targeted the attention domain.

We aimed to evaluate:

1. The prevalence and patterns of VADs in PWA following chronic left hemisphere strokes.
2. The effect of cholinergic modulation (donepezil) on VADs, in a proof-of-concept randomized, placebo-controlled clinical trial.

STUDY 1: Comparison of outcomes on the CalCAP⁸ from 55 PWA^{9,10,11} relative to 25 healthy controls and a normative dataset (<2 SDs). Mann-Whitney U tests were conducted to compare reaction time (RT) and accuracy between groups.

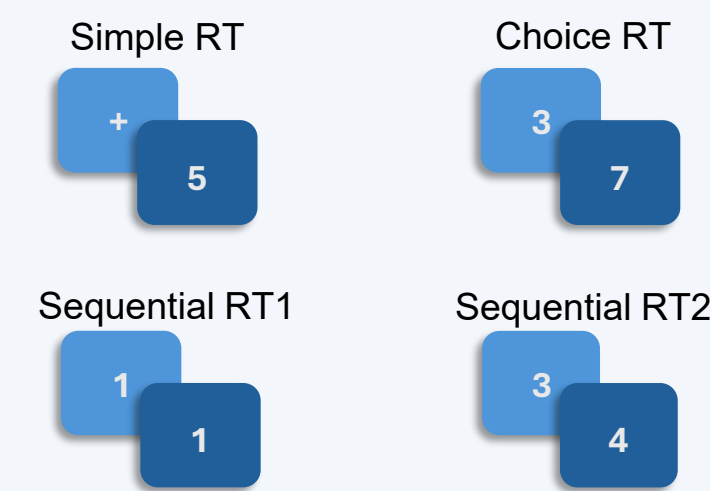
STUDY 2: Proof-of-concept double-blind, parallel, randomized, placebo-controlled clinical trial of the cholinergic agent donepezil⁹ (ClinicalTrials.gov ID: NCT00196690). 2 × 2 repeated-measures ANOVA were performed compare pre- and post-treatment CalCAP and language outcomes.

Language assessment:

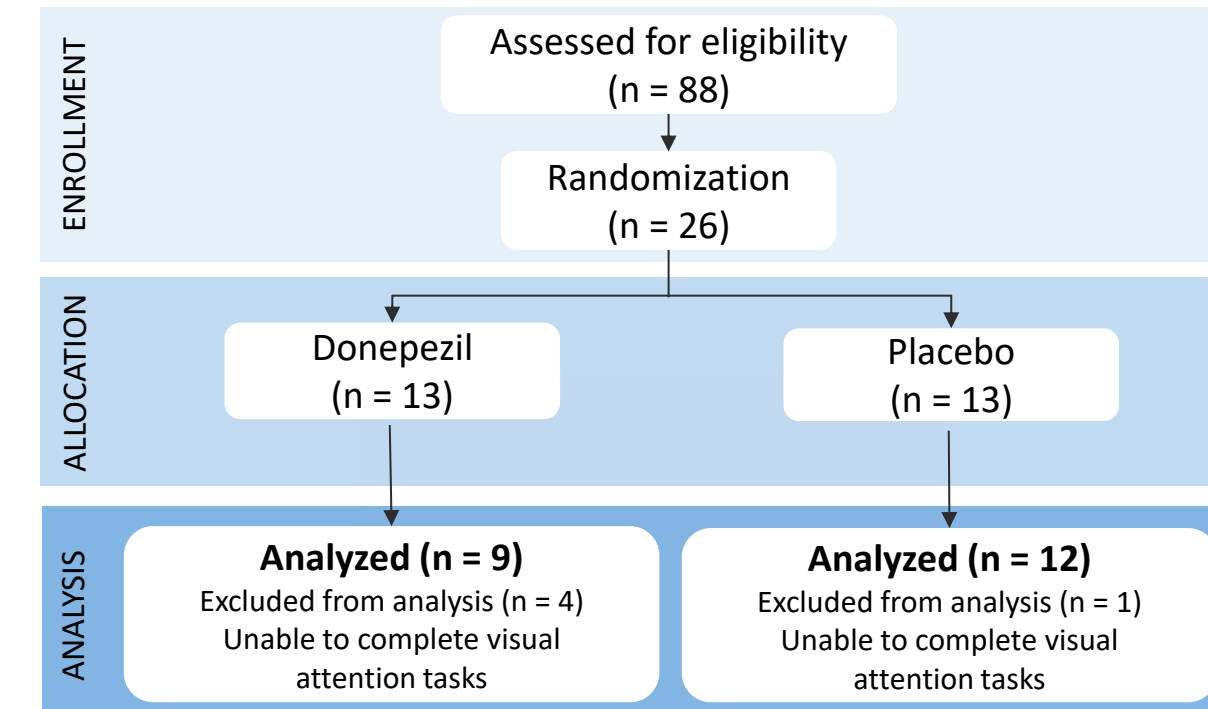
In both studies, aphasia was evaluated with the Western Aphasia Battery (WAB).

METHODS

California Computerized Attention Package (CalCAP) Abbreviated



STUDY 2 CONSORT DIAGRAM



RESULTS

STUDY 1: VADs prevalence among PWA

- There was a high occurrence of VADs among the PWA group (Figure 1).
- PWA were slower and less accurate than healthy controls in all tasks (Figure 2).
- Among PWA with abnormal attention, outcomes in the most demanding task (Sequential RT2) correlated with aphasia severity ($r = .413$, $p < .05$).

Figure 1. VADs impairment patterns

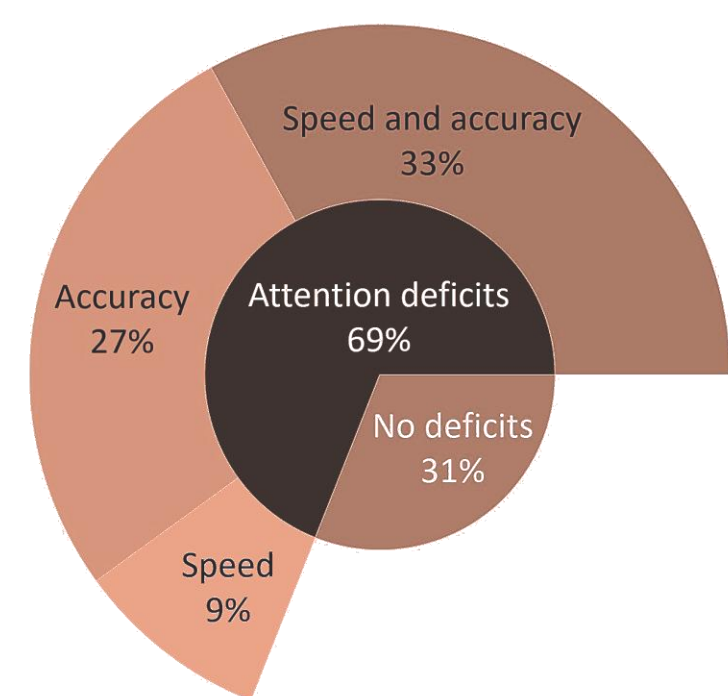
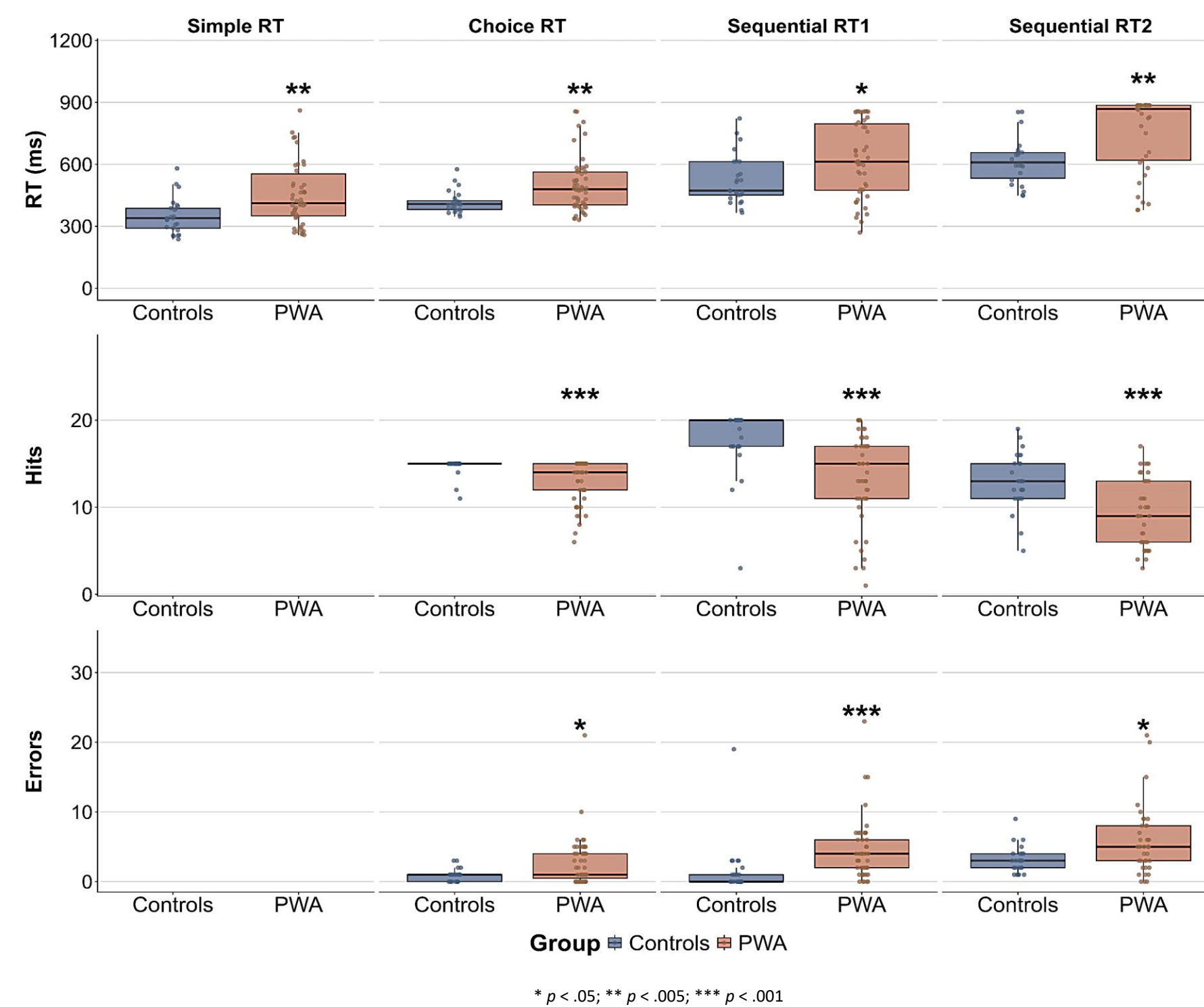


Figure 2. PWA vs Controls CalCAP scores¹²



STUDY 2: cholinergic modulation of VADs

Figure 3. Donepezil vs Placebo CalCAP scores¹²

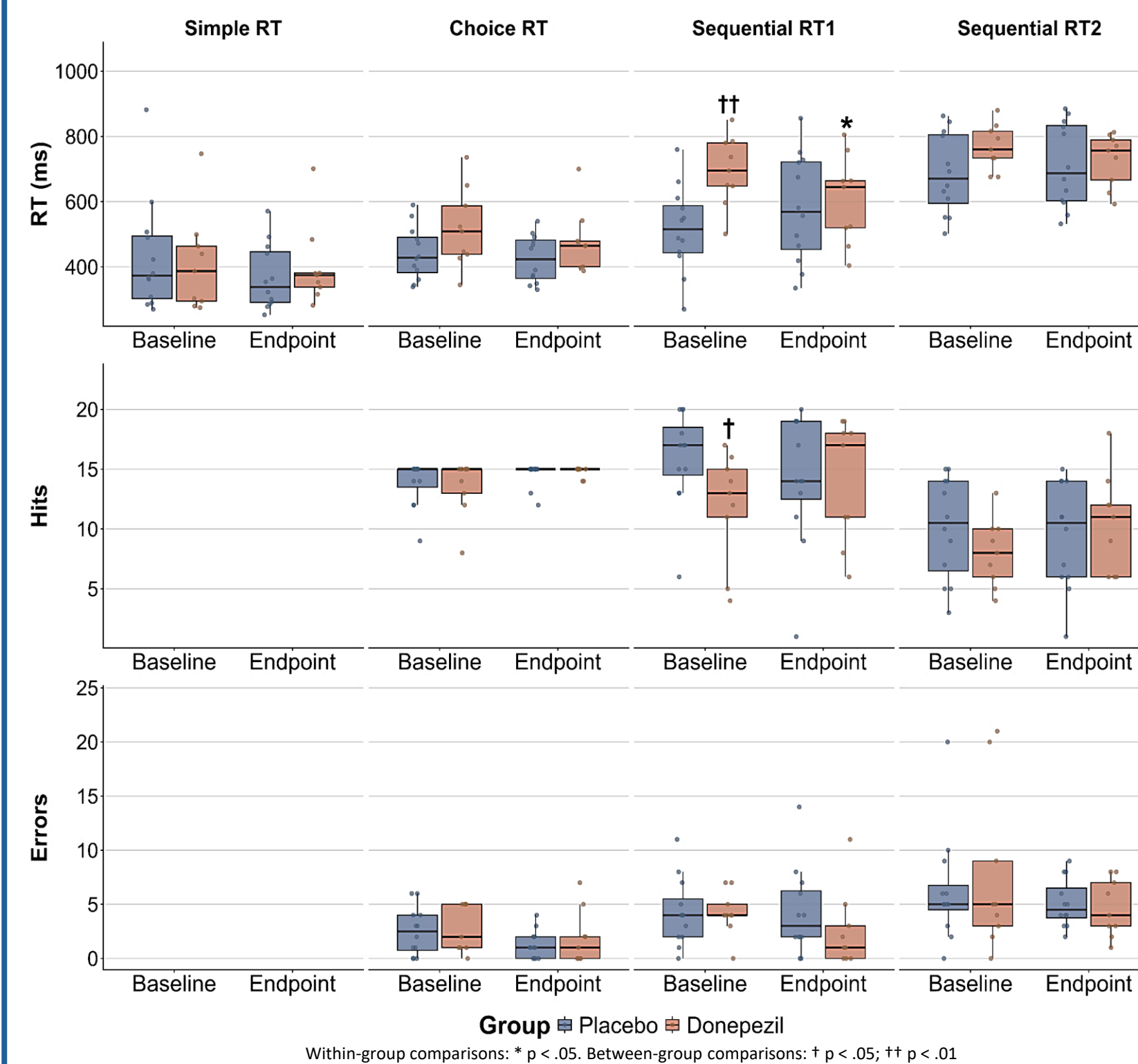
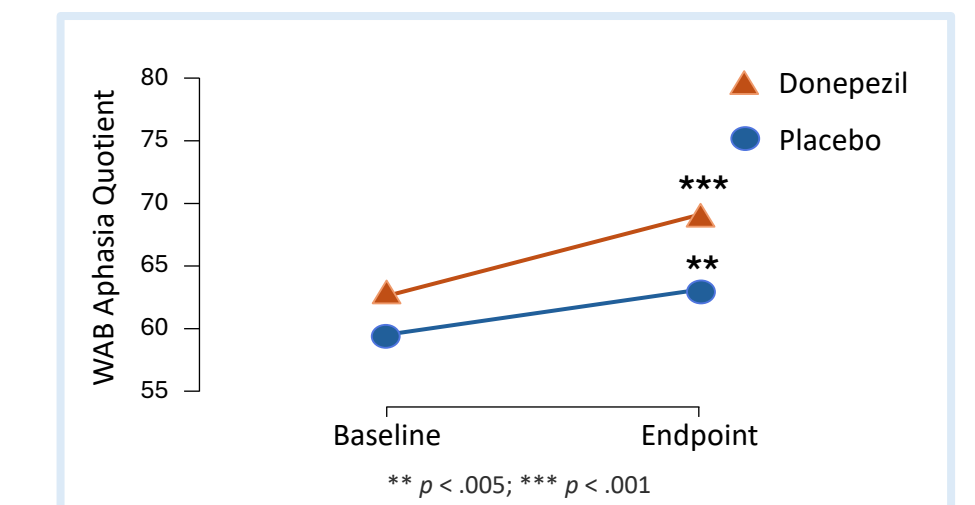


Figure 4. Donepezil vs Placebo WAB-AQ scores



- The donepezil group significantly improved speed in the Sequential RT1 task ($p = 0.032$) and showed a tendency toward improvement in the Sequential RT2 task, the two most demanding tasks (Figure 3).
- There was a significant group × time interaction ($p = .012$): Δ Donepezil = 7.34 ($p < .001$); Δ Placebo = 3.33 ($p = .001$) (Figure 4).
- Baseline aphasia severity significantly correlated with gains in Sequential RT2 ($r = .503$; $p = .02$).

CONCLUSIONS

- VADs are highly prevalent in PWA, linked to aphasia severity, and partly addressable via cholinergic stimulation.
- These results open new treatment pathways for non-verbal deficits in aphasia.

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